

SOLAR-POWERED GENERATORS: Reduce Pollution And Downtime

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Solar-powered generators are noiseless, environment-safe and can bring down the usage of exhaustible fuels drastically

Offices, factories, residential complexes and critical units (like hospitals and data centres), among others, in India require backup power generators. These are needed to fight the inconsistent supply of electricity that threatens to thwart essential operations. Generators mainly run on fossil fuels such as diesel or petrol. These emit huge amounts of smoke, heat and sound. This has caused much harm to the environment, and led to various restrictions being enforced on the use of generators in major cities.

Solar energy, in the form of solar-powered generators, can come to aid in this hour of need. These devices can drastically bring down the usage of fuels as well as the carbon emission rate. In addition, there are many long-term financial benefits.

Advantages of green generators

Worsening environmental conditions are attributed to excessive carbon and green-

house gases emissions, pollutant particulate matters (PM 2.5 and PM 10) and other harmful elements being added to the air. A large part of this is produced by diesel generator sets, as is evident from a recent report by Centre for Science and Environment.

The report reveals that a city like Gurugram is facing a spike of at least 30 per cent in air pollution caused by diesel generator sets. PM 2.5 and PM 10 level readings have increased variably between 30 and 100 per cent.

Results are similar in other regions, too. The government is now pushing Indian consumers to install solar solutions to leverage India's abundant sunlight availability. Solar-powered generator sets is one such useful solution.

Arvind Patel, executive director, Divya Hitech Solar, says, "Most traditional generators are very noisy. These run completely on diesel (or petrol), consuming exhaustible resources.



Fig. 1: Green generator by Maru Systems; outside view (left) and inside view (right) (Credit: divyahitechsolar.com)

